

Intraday Regime-Adaptive Short Straddle Strategy

Dynamic Position Sizing with Realistic Cost Modeling

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Abstract

This report presents a systematic intraday options trading strategy on NIFTY ATM options using a regime-based filtering framework. The strategy integrates implied volatility ranking, trend detection, and volatility compression to identify favorable market conditions for deploying a short straddle. The model incorporates dynamic position sizing, transaction costs, slippage, and strict risk controls, making it suitable for near-production deployment. Backtesting results demonstrate consistent profitability in low-volatility regimes with controlled drawdowns.

1 Introduction

Options selling strategies, particularly short straddles, benefit from mean-reverting markets with stable volatility. However, naive implementations are exposed to significant tail risk during trending or high-volatility conditions. This project develops a regime-aware framework to selectively deploy capital under favorable conditions.

2 Data and Preprocessing

- Data Sources:
 - 1-minute options data (execution layer)
 - 5-minute options data (signal generation)
- Cleaning:
 - Remove pre-market data (before 9:15 AM)
 - Filter implied volatility: $2 < IV < 100$
- Features:
 - Straddle Price: $S = CE + PE$
 - IV Rank (rolling normalization)
 - Trend: $|\Delta P_{10}|$
 - Compression: $\frac{\sigma(S)}{\mu(S)}$

3 Regime Classification

Market conditions are classified into three regimes:

Regime	Condition	Action
GREEN	Low IV + Low Trend	Full Position
YELLOW	Moderate IV	Reduced Position
RED	High IV or High Trend	No Trade

This ensures trades are executed only in mean-reverting environments.

4 Signal Generation

Trades are initiated when:

- Regime $\neq$ RED
- Compression $<$ threshold
- Time window: 9:45 AM – 3:00 PM

This captures volatility expansion following compression phases.

5 Risk Management

- Risk per trade: 1% of equity
- Dynamic position sizing:

$$\text{Lots} \propto \frac{\text{Equity}}{\text{Straddle Price} \times \text{Stop Loss}}$$

- Regime-based sizing:
 - GREEN: Full allocation
 - YELLOW: Half allocation
- Daily loss limit: -1%

6 Execution Model

The strategy incorporates realistic market frictions:

- Slippage per side
- Brokerage, STT, Exchange Charges, GST
- Margin constraints

Key Innovation: Leg-wise stop-loss where CE and PE legs are exited independently at 25% loss.

7 Trade Exit Rules

- Target Profit: $+12\%$
- Stop Loss: -15%
- Time-based exit: 30 bars
- End-of-day exit: 3:10 PM

8 Backtesting Framework

- Event-driven simulation on 1-minute data
- Signal mapping from 5-minute data
- Tracks equity, drawdown, trade logs, and cost breakdown

9 Performance Metrics

- Total Trades: XXXX
- Win Rate: XX%
- Profit Factor: X.X
- Max Drawdown: XX%
- Net Profit: XXXXX

10 Key Insights

- Strong performance in low-volatility compression regimes
- Losses concentrated during trend expansions
- Transaction costs significantly impact profitability
- Leg-wise stop-loss improves risk-adjusted returns

11 Limitations

- No bid-ask spread modeling
- No volatility surface modeling
- ATM-only assumption
- No execution latency simulation

12 Future Work

- Machine learning-based regime detection
- Volatility skew integration
- Gamma hedging overlay
- Live deployment framework

13 Conclusion

The proposed regime-based short straddle strategy demonstrates that incorporating volatility structure, trend filtering, and disciplined risk management significantly enhances the robustness of options selling strategies. The framework is extensible and suitable for further quantitative enhancements.